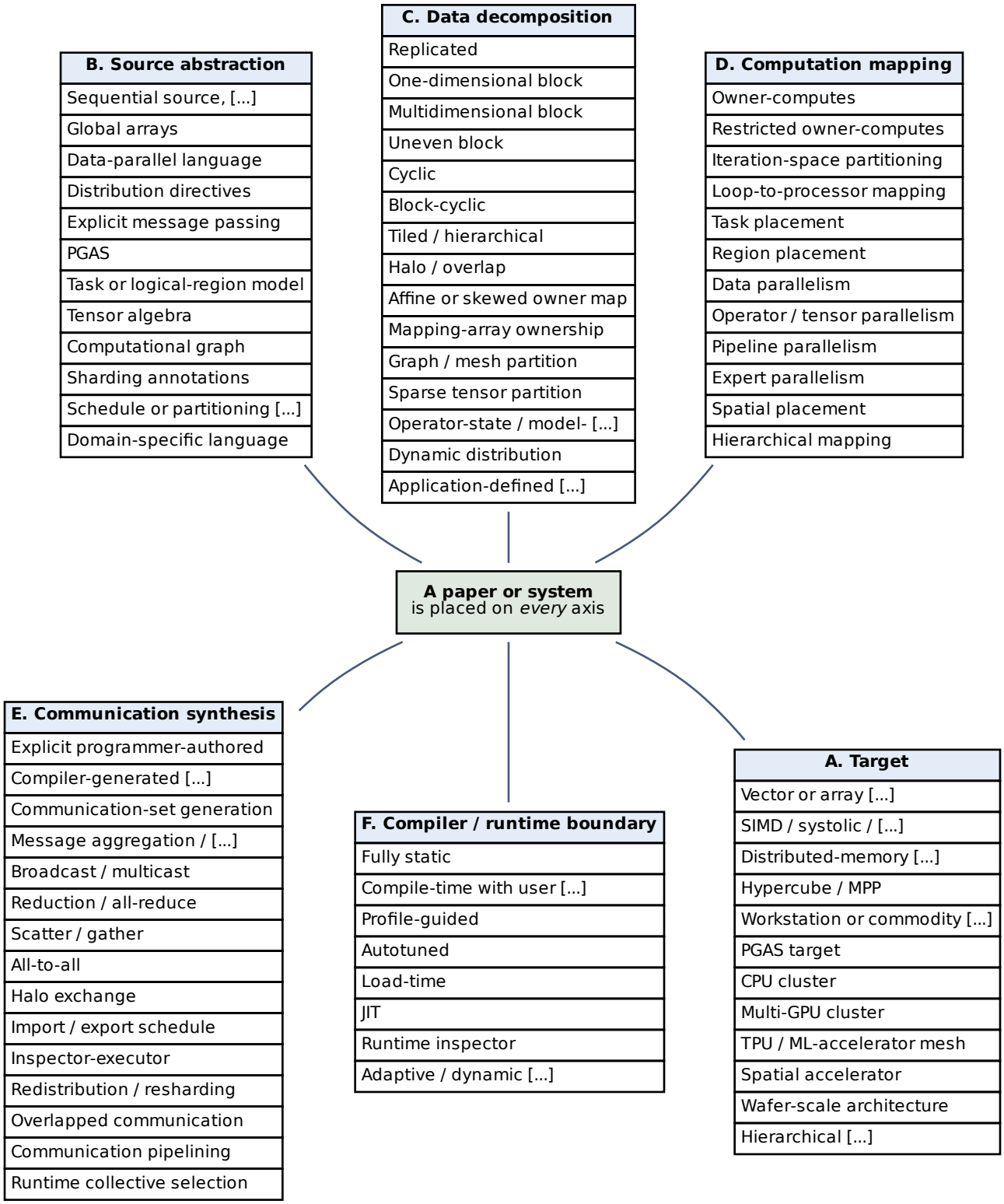
B. Source abstraction
Sequential source, [...]
Global arrays
Data-parallel language
Distribution directives
Explicit message passing
PGAS
Task or logical-region model
Tensor algebra
Computational graph
Sharding annotations
Schedule or partitioning [...]
Domain-specific language
C. Data decomposition
Replicated
One-dimensional block
Multidimensional block
Uneven block
Cyclic
Block-cyclic
Tiled / hierarchical
Halo / overlap
Affine or skewed owner map
Mapping-array ownership
Graph / mesh partition
Sparse tensor partition
Operator-state / model- [...]
Dynamic distribution
Application-defined [...]
D. Computation mapping
Owner-computes
Restricted owner-computes
Iteration-space partitioning
Loop-to-processor mapping
Task placement
Region placement
Data parallelism
Operator / tensor parallelism
Pipeline parallelism
Expert parallelism
Spatial placement
Hierarchical mapping
A paper or system
is placed on every axis
E. Communication synthesis
Explicit programmer-authored
Compiler-generated [...]
Communication-set generation
Message aggregation / [...]
Broadcast / multicast
Reduction / all-reduce
Scatter / gather
All-to-all
Halo exchange
Import / export schedule
Inspector-executor
Redistribution / resharding
Overlapped communication
Communication pipelining
Runtime collective selection
F. Compiler / runtime boundary
Fully static
Compile-time with user [...]
Profile-guided
Autotuned
Load-time
JIT
Runtime inspector
Adaptive / dynamic [...]
A. Target
Vector or array [...]
SIMD / systolic / [...]
Distributed-memory [...]
Hypercube / MPP
Workstation or commodity [...]
PGAS target
CPU cluster
Multi-GPU cluster
TPU / ML-accelerator mesh
Spatial accelerator
Wafer-scale architecture
Hierarchical [...]